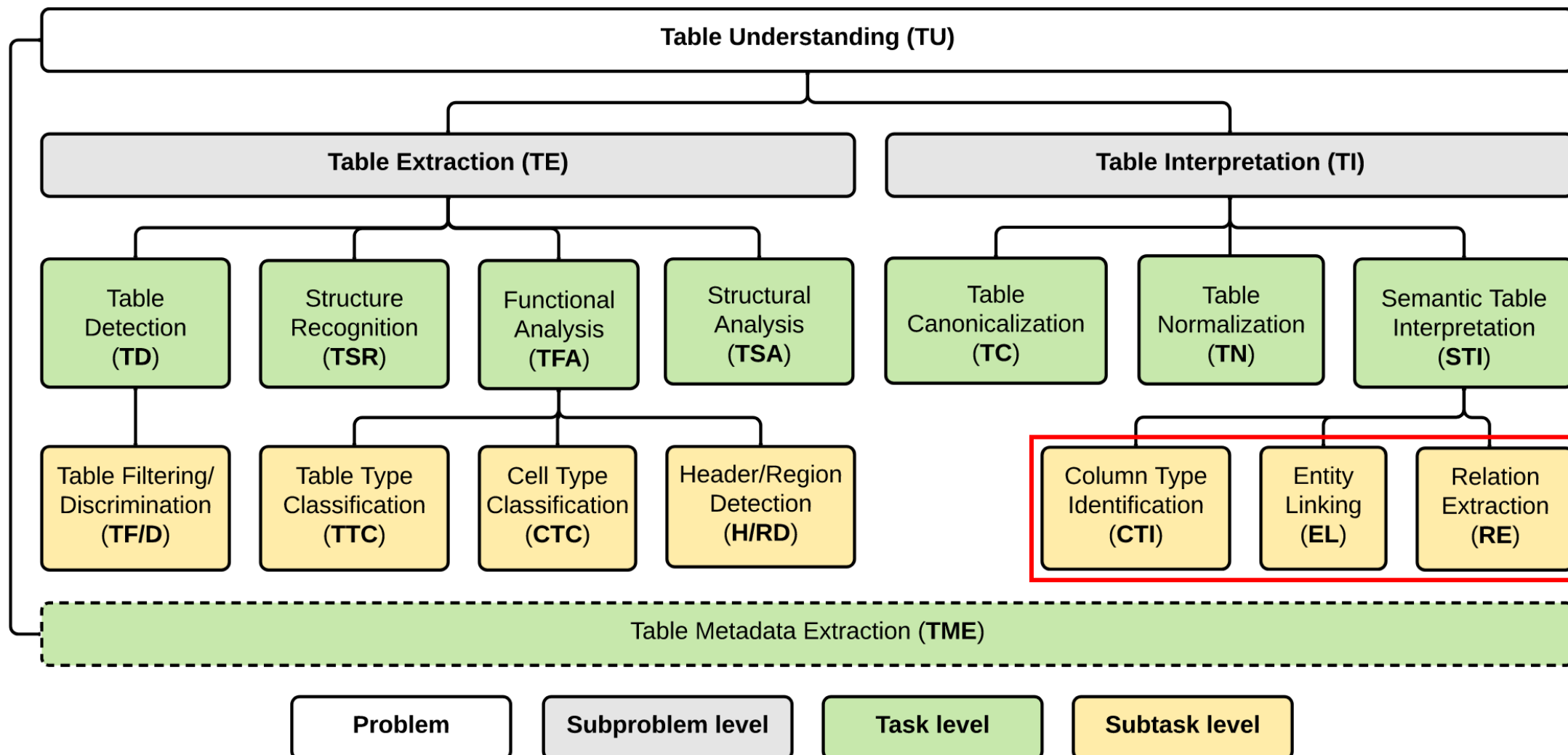


# TURL: Table Understanding through Representation Learning

VLDB,2020 (SIGMOD,2022)

The Ohio State University & Google Research

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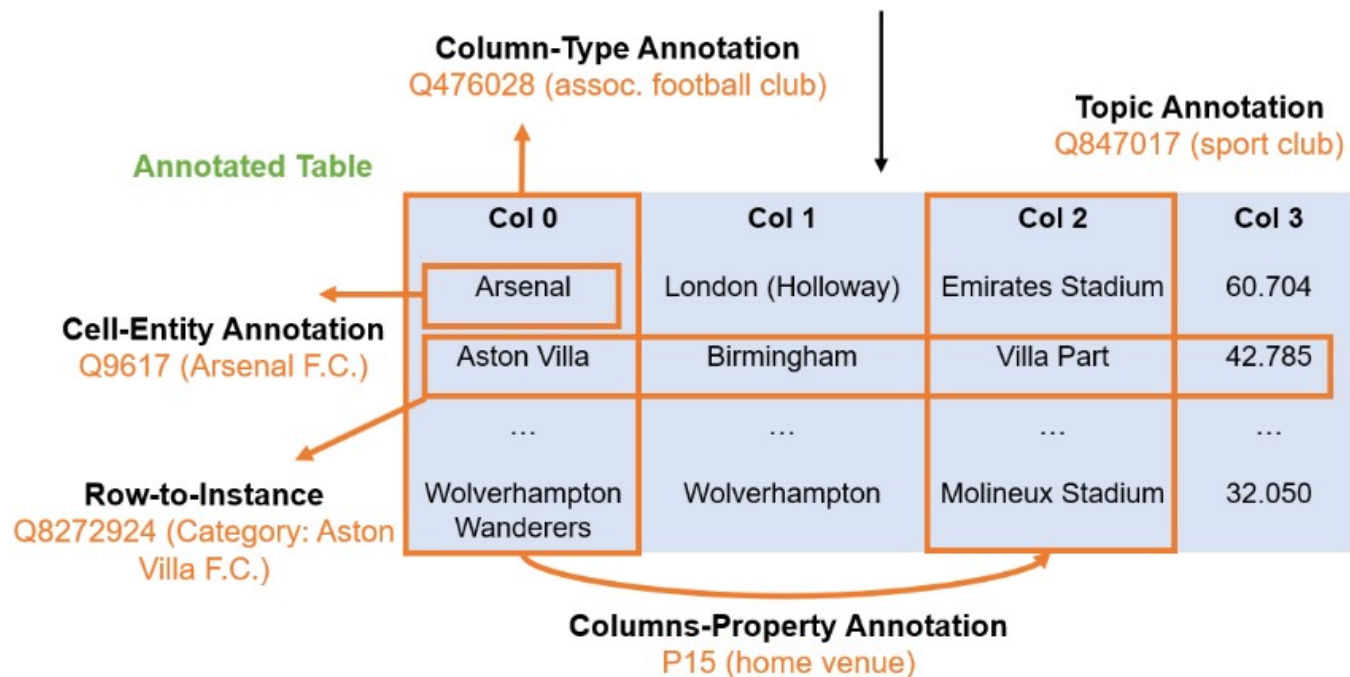


(Table understanding: Problem overview )

Input Table

| Col 0                   | Col 1             | Col 2            | Col 3  |
|-------------------------|-------------------|------------------|--------|
| Arsenal                 | London (Holloway) | Emirates Stadium | 60.704 |
| Aston Villa             | Birmingham        | Villa Part       | 42.785 |
| ...                     | ...               | ...              | ...    |
| Wolverhampton Wanderers | Wolverhampton     | Molineux Stadium | 32.050 |

Annotated Table



(Towards Generative Semantic Table Interpretation)

表格理解目的在于，通过对非结构化或半结构化的表格进行 CEA、CPA、CTA 等操作，提取、挖掘表格潜在的语义信息，再用作 KG 扩充等后续处理。

CEA:

将表格内实体提及连接到外部知识库。

CTA:

根据该列所包含的 entity，确定列类型。

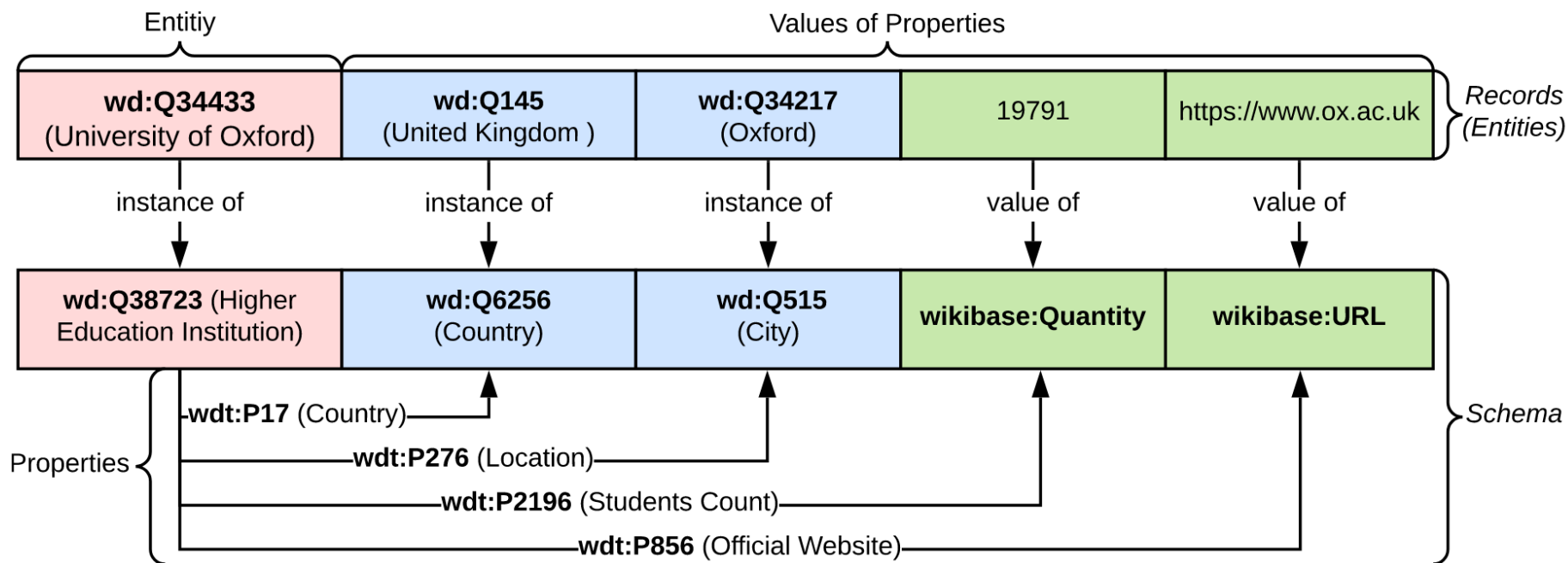
CPA:

给定一对 subject-object column，确定它们之间的关系（KB 中存在），结果返回 RDF 三元组。

| Entity Column        |         | Property Columns |                  |                                                               |
|----------------------|---------|------------------|------------------|---------------------------------------------------------------|
| Organization         | Country | City             | Num. of Students | URL                                                           |
| University of Oxford | UK      | Oxford           | 19,791           | <a href="https://www.ox.ac.uk">https://www.ox.ac.uk</a>       |
| University of London | UK      | London           | 45,500           | <a href="https://london.ac.uk">https://london.ac.uk</a>       |
| Cornell University   | US      | Ithaca           | 23,620           | <a href="https://www.cornell.edu">https://www.cornell.edu</a> |

Mentions of Entities
Literals

(a)



(b)

## National Film Award for Best Direction

From Wikipedia, the free encyclopedia

page title &  
topic entity

### Winners [\[edit\]](#)

section title

List of award recipients, showing the year, film and language

caption

| Year <sup>[b]</sup> | Recipient    | Film                         | Language | Ref  |
|---------------------|--------------|------------------------------|----------|------|
| 1967<br>(15th)      | Satyajit Ray | <i>Chiriyakhana</i>          | Bengali  | [13] |
| 1968<br>(16th)      | Satyajit Ray | <i>Goopy Gyne Bagha Byne</i> | Bengali  | [14] |
| 1969<br>(17th)      | Mrinal Sen   | <i>Bhuvan Shome</i>          | Hindi    | [15] |
| 1970<br>(18th)      | Satyajit Ray | <i>Pratidwandi</i>           | Bengali  | [16] |

headers

entity

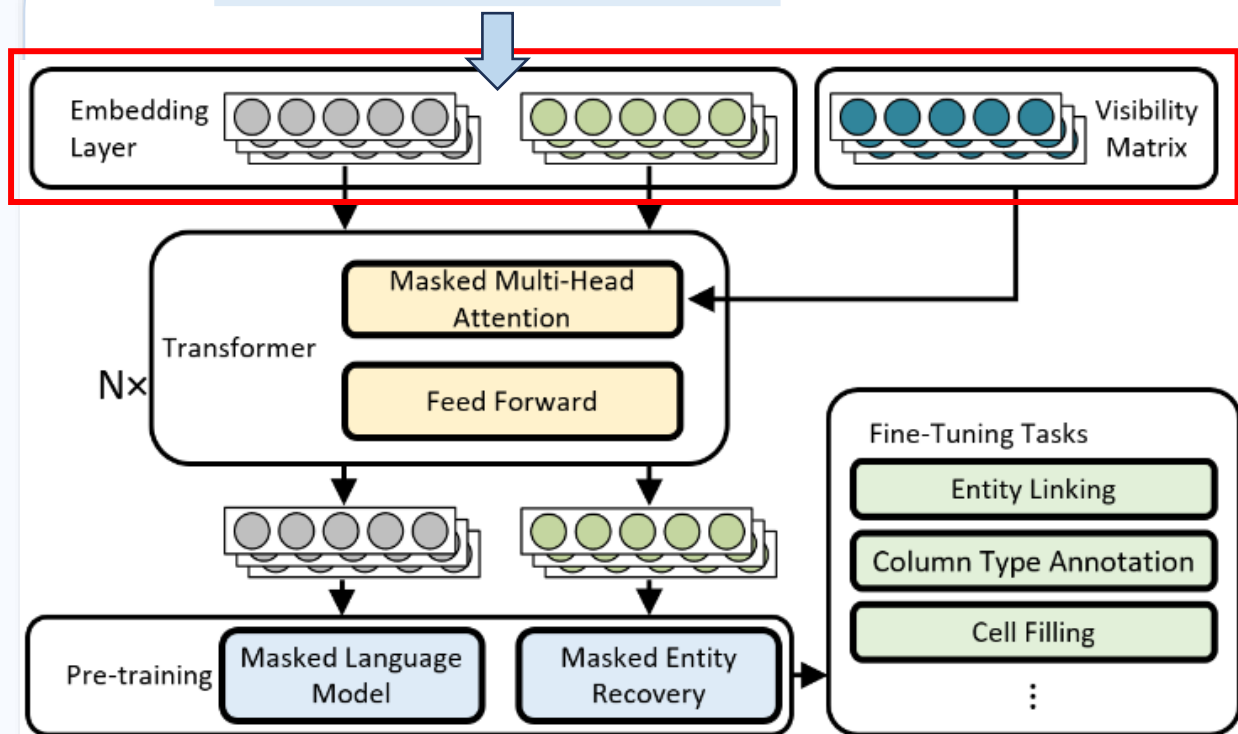
object  
columns

subject column (year here are linked to specific events)

(Relational table from Wikipedia)

本文认为，以往对于 TI 任务的处理方法过于繁琐且特殊化，不具有泛化性。BERT 的 Pre-Training/Fine-Tuning 模式对于无监督文本（无结构）分析有良好的效果，本文尝试构建一种基于该模式的 TURL 框架，将其应用于对关系表格（半结构或结构化）的理解。

Input token: 标题、主题、表头等  
Input entity: 包含实体的单元格



## Embedding Layer:

Input token representation:

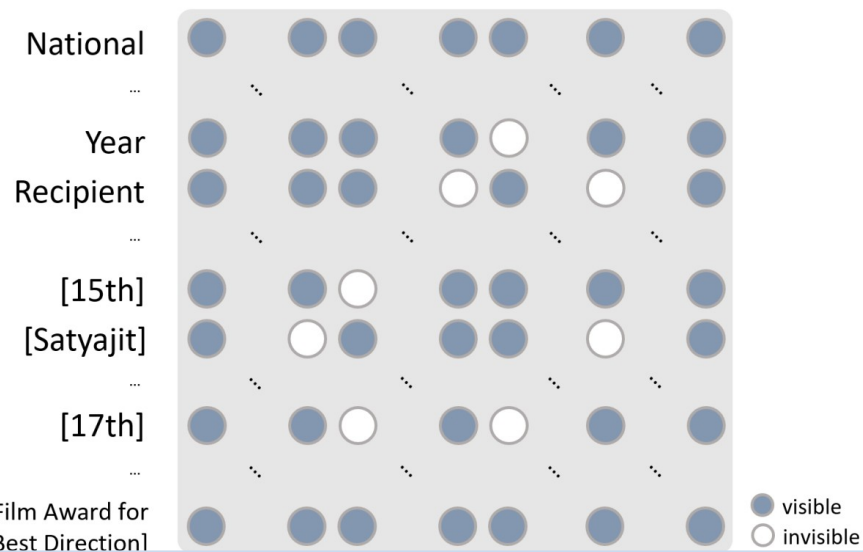
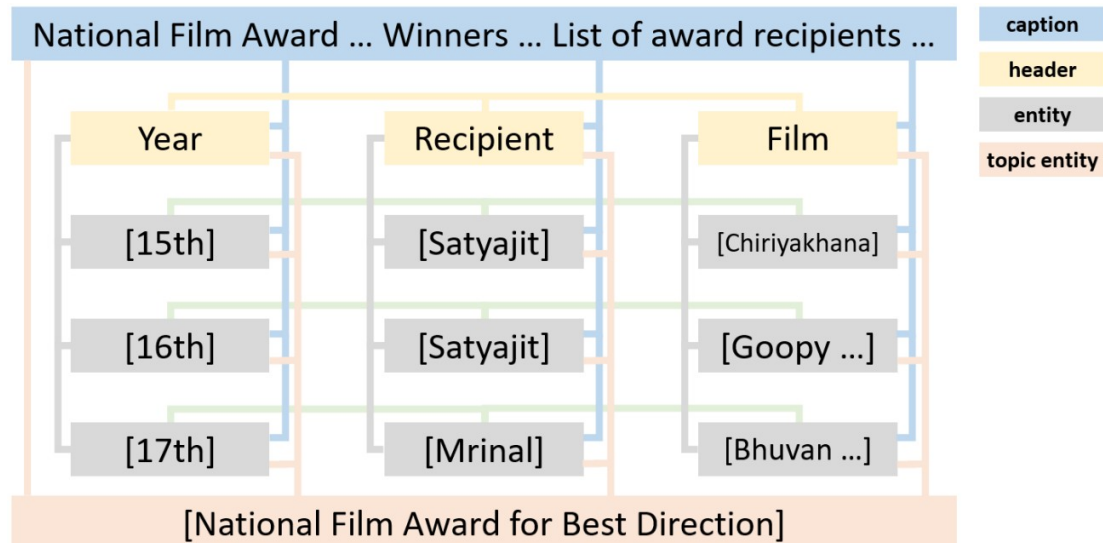
$$\mathbf{x}^t = \mathbf{w} + \mathbf{t} + \mathbf{p}$$

Input entity representation:

$$\mathbf{x}^e = \text{LINEAR}([\mathbf{e}^e; \mathbf{e}^m]) + \mathbf{t}^e;$$

$$\mathbf{e}^m = \text{MEAN}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_j, \dots)$$

| Symbol         | Description       |
|----------------|-------------------|
| $\mathbf{w}$   | 词嵌入向量             |
| $\mathbf{t}$   | token 类型          |
| $\mathbf{p}$   | token 位置          |
| $\mathbf{e}^e$ | the linked entity |
| $\mathbf{e}^m$ | entity mention    |
| $\mathbf{t}^e$ | 实体类型              |



Structure-aware Transformer:

本质上就是 Transformer-encoder，引入多头注意力机制，区别在于做 MASK 时，引入一个 Visibility Matrix (symmetric)：

$$\text{MultiHead}(\mathbf{h}) = [\text{head}_1; \dots; \text{head}_i; \dots; \text{head}_k] W^O;$$

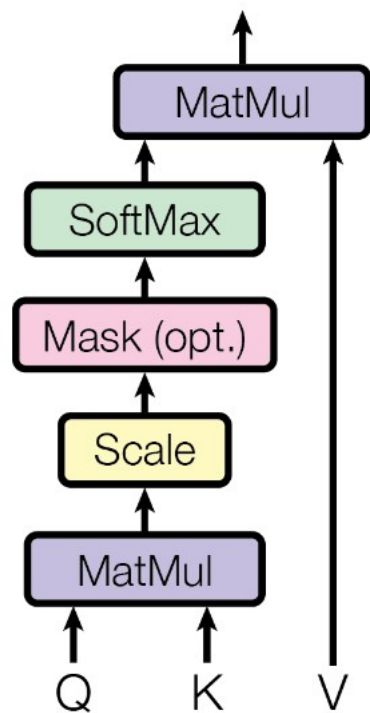
$$\text{head}_i = \text{Attention}(\mathbf{h} W_i^Q, \mathbf{h} W_i^K, \mathbf{h} W_i^V);$$

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} M\right) V.$$

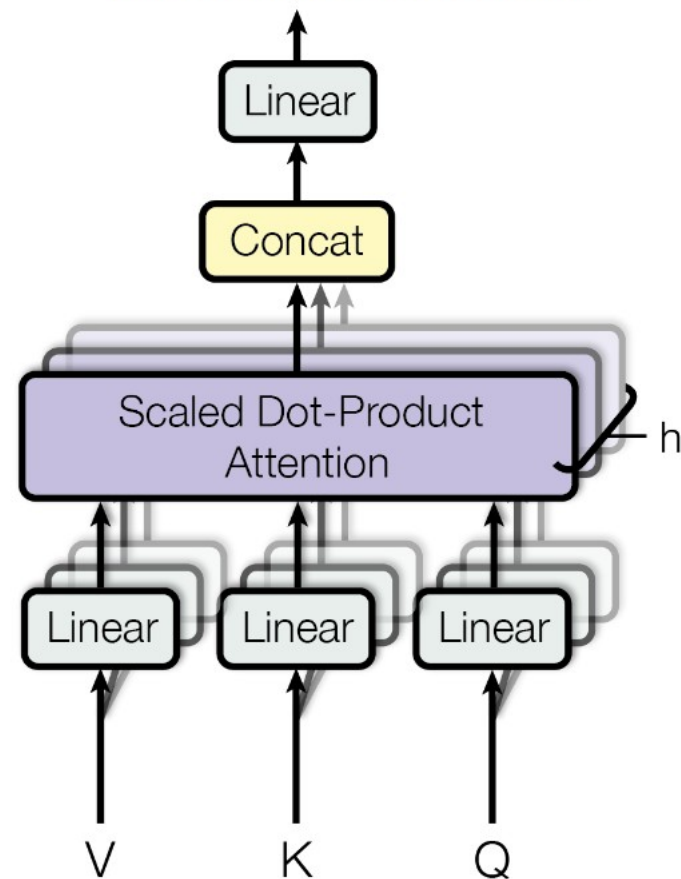
Visibility Matrix 定义：

- 表格标题和主题实体对所有实体可见，对应位置记 1；
- 处于同行同列的实体互相均可见，对应位置记 1。

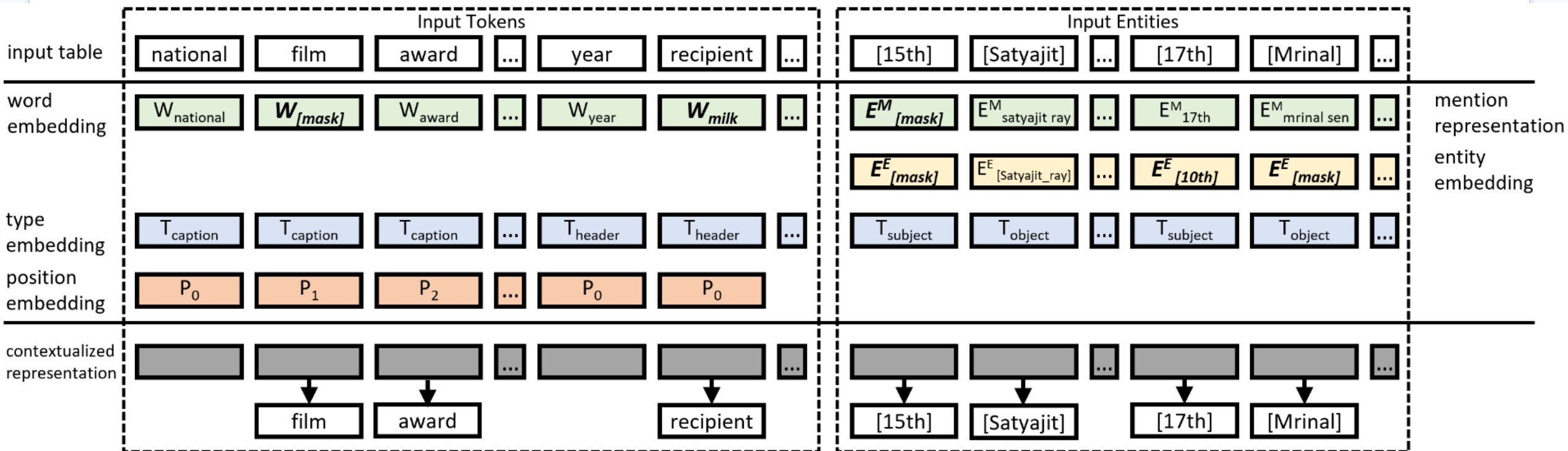
## Scaled Dot-Product Attention



## Multi-Head Attention



(Attention is All you Need)



## Pre-Training

### MLM (Masked Language Model)

$$P(w) = \frac{\exp(\text{LINEAR}(h^t) \cdot w)}{\sum_{w_k \in \mathcal{W}} \exp(\text{LINEAR}(h^t) \cdot w_k)}$$

### MER (Masked Entity Recovery)

$$P(e) = \frac{\exp(\text{LINEAR}(h^e) \cdot e^e)}{\sum_{e_k \in \mathcal{E}} \exp(\text{LINEAR}(h^e) \cdot e_k^e)}$$

| Symbol | Description             |
|--------|-------------------------|
| $w$    | token 词嵌入向量             |
| $e^e$  | the linked entity 词嵌入向量 |
| $h^t$  | token 上下文表征             |
| $h^e$  | entity 上下文表征            |

Pre-Training 直接使用 BERT 的 MLM 模块，在 Transformer-encoder 后接 MLP 输出预测结果，需要提供词嵌入向量、对应的上下文表征、Masked 词汇位置。MER 类似。

## Data Pre-Processing

- 将上文提到的 token 整合为综合性描述
- 确保语料表格规模，多列多行
- 规范化实体提及，实体链接只取第一个结果
- 剔除包含合并列的行

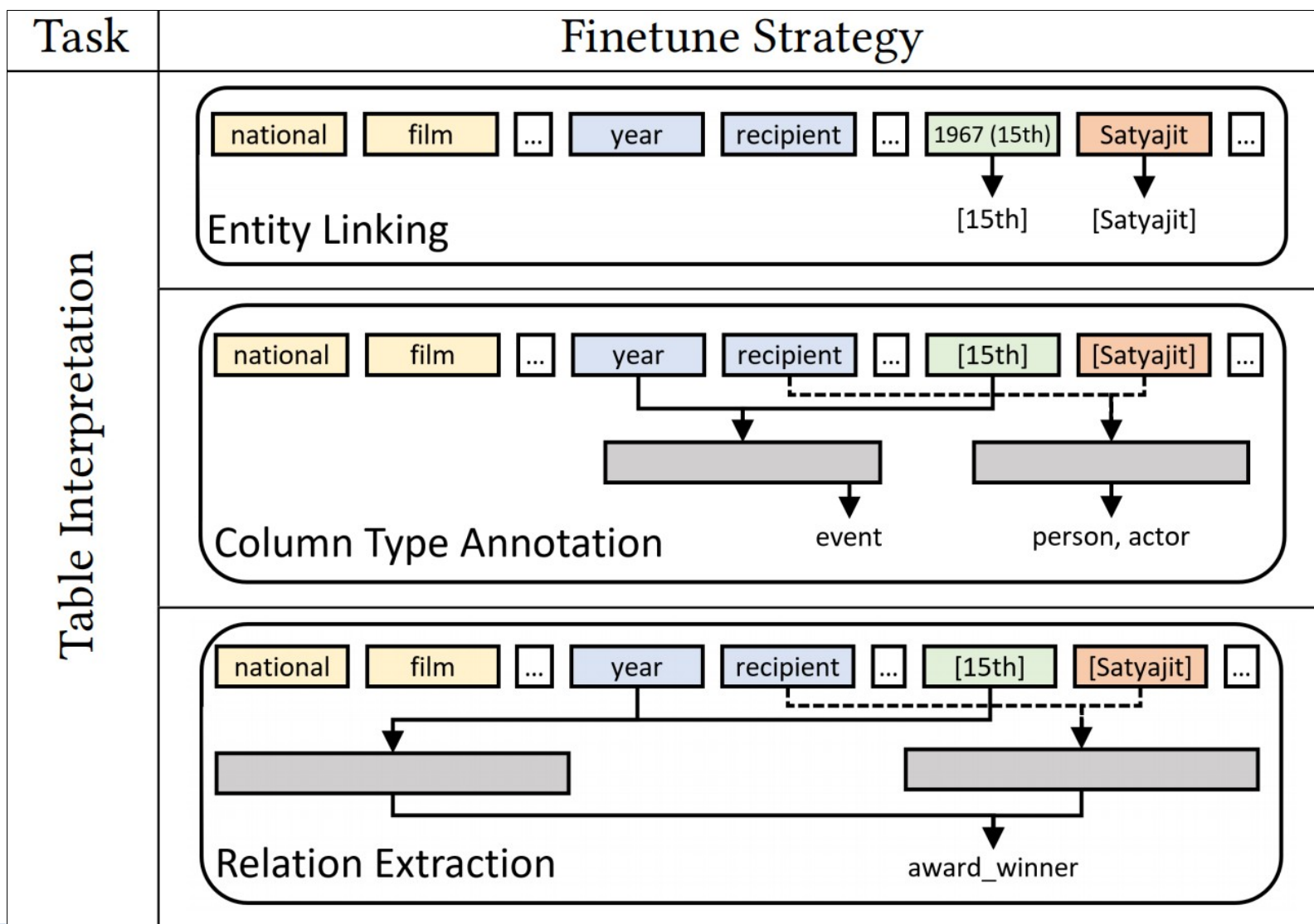
## Identify relational tables

- 定位至少包含一个 entity 的列，并过滤噪声列
- 识别关系表，必须包含主列 subject column
- 剔除包含少于三个 entities 或 20 列以上的关系表

## Data partitioning

筛选要求：

- 主列包含四个以上 linked entities
- 至少三个实体列（含主列）
- 实体列中一半以上的单元格有 linked entity
- 570171 / 5036 / 4964



Fine-Tuning:

$$\mathbf{e}^{kb} = [\text{MEAN}_{w \in N}(\mathbf{w}), \text{MEAN}_{w \in D}(\mathbf{w}), \text{MEAN}_{t \in T}(\mathbf{t})]$$

同包含 matedata、cell text 的  $\mathbf{h}^e$  计算相似度, 见前文

| Symbol         | Description           |
|----------------|-----------------------|
| $w$            | 实体词嵌入向量               |
| $N$            | 实体名称 (from Wikipedia) |
| $D$            | 实体描述 (from Wikipedia) |
| $T$            | 实体类型 (from DBpedia)   |
| $\mathbf{h}^e$ | entity 上下文表征          |

Baseline:

Task-specific Datasets

| Method              | WikiGS    |           |           | Our Test Set |           |           | T2D       |           |           |
|---------------------|-----------|-----------|-----------|--------------|-----------|-----------|-----------|-----------|-----------|
|                     | F1        | P         | R         | F1           | P         | R         | F1        | P         | R         |
| T2K [36]            | 34        | 70        | 22        | -            | -         | -         | 82        | <b>90</b> | 76        |
| Hybrid II [17]      | 64        | 69        | <b>60</b> | -            | -         | -         | <b>83</b> | 85        | <b>81</b> |
| Wikidata Lookup     | 57        | 67        | 49        | 62           | 62        | 60        | 80        | 86        | 75        |
| TURL + fine-tuning  | <b>67</b> | <b>79</b> | 58        | <b>68</b>    | <b>71</b> | <b>66</b> | 78        | 83        | 73        |
| w/o entity desc.    | 60        | 70        | 52        | 60           | 63        | 58        | -         | -         | -         |
| w/o entity type     | 66        | 78        | 57        | 67           | 70        | 65        | -         | -         | -         |
| + reweighting       | -         | -         | -         | -            | -         | -         | 82        | 88        | 77        |
| WikiLookup (Oracle) | 74        | 88        | 64        | 79           | 82        | 76        | 90        | 96        | 84        |

Fine-Tuning:

$$\mathbf{h}_c = [\text{MEAN}(\mathbf{h}_i^t, \dots); \text{MEAN}(\mathbf{h}_j^e, \dots)]$$

$$P(l) = \text{Sigmoid}(\mathbf{h}_c W_l + b_l)$$

| Symbol         | Description          |
|----------------|----------------------|
| <b>Symbol</b>  | 列中实体类型 (集合)          |
| $\mathbf{h}_c$ | 列表征                  |
| $\mathbf{h}^t$ | tokens in header 的表征 |
| $\mathbf{h}^e$ | entity cells 的表征     |

Task-specific Datasets:

从 Freebase 获取候选类型集, 只保留具有代表性的类型, 剔除少于 100 例的类型, 测试在 T2D, Eftymiou(有标注的 WikiGS 子集) 小规模数据集, 并进行一些小规模的人工类型映射。

Baseline:

| Method                                   | F1           | P            | R            |
|------------------------------------------|--------------|--------------|--------------|
| Sherlock (only entity mention) [22]      | 78.47        | 88.40        | 70.55        |
| TURL + fine-tuning (only entity mention) | 88.86        | 90.54        | 87.23        |
| TURL + fine-tuning                       | <b>94.75</b> | <b>94.95</b> | <b>94.56</b> |
| w/o table metadata                       | 93.77        | 94.80        | 92.76        |
| w/o learned embedding                    | 92.69        | 92.75        | 92.63        |
| only table metadata                      | 90.24        | 89.91        | 90.58        |
| only learned embedding                   | 93.33        | 94.72        | 91.97        |

| Method                                   | T2D-Te | Eftymiou |
|------------------------------------------|--------|----------|
| HNN + P2Vec (entity mention + KB) [11]   | 0.966  | 0.865    |
| TURL + fine-tuning (only entity mention) | 0.888  | 0.745    |
| + table metadata                         | 0.860  | 0.904    |

| Method                 | person       | pro_athlete  | actor        |
|------------------------|--------------|--------------|--------------|
| Sherlock [22]          | 96.85        | 74.39        | 29.07        |
| TURL + fine-tuning     | <b>99.71</b> | <b>91.14</b> | <b>74.85</b> |
| only entity mention    | 98.44        | 87.11        | 58.86        |
| only table metadata    | 98.26        | 88.80        | 70.86        |
| only learned embedding | 98.72        | 91.06        | 73.62        |

Fine-Tuning:

$$P(r) = \text{Sigmoid}([\mathbf{h}_c; \mathbf{h}_{c'}]W_r + b_r)$$

| Symbol            | Description   |
|-------------------|---------------|
| Symbol            | KB 中存在的关系实体   |
| $\mathbf{h}_c$    | 列表征           |
| $\mathbf{h}_{c'}$ | pair 中的另一个列表征 |

Task-specific Datasets:

从 Freebase 获取候选关系集，半数以上 entity pair 拥有的关系才被选用，且被选中的关系要有至少 100 例数据。

Baseline:

| Method                                   | F1           | P            | R            |
|------------------------------------------|--------------|--------------|--------------|
| BERT-based                               | 90.94        | 91.18        | 90.69        |
| TURL + fine-tuning (only table metadata) | 92.13        | 91.17        | 93.12        |
| TURL + fine-tuning                       | <b>94.91</b> | <b>94.57</b> | <b>95.25</b> |
| w/o table metadata                       | 93.85        | 93.78        | 93.91        |
| w/o learned embedding                    | 93.35        | 92.90        | 93.80        |

